

Student Number: \_\_\_\_\_



**ROSEVILLE COLLEGE**

**MATHEMATICS**

**2003**

**PRELIMINARY EXAMINATION**

## General Instructions

**Time Allowed – 3 hours, plus 5 minutes reading time**

**Directions to Candidates:**

- Attempt all questions.
- Begin each question on a new page, with your number clearly written on every page.
- There are 10 questions, each worth 12 marks.
- Individual marks for each question are indicated on the left hand side of each question
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.

**Question 1**

**Basic Arithmetic** (*Start work on a new page*)

(a) Simplify  $|-5| - |8|$  1

(b) Calculate correct to two decimal places 2

$$\frac{34.2}{16.3 \times 2.7}$$

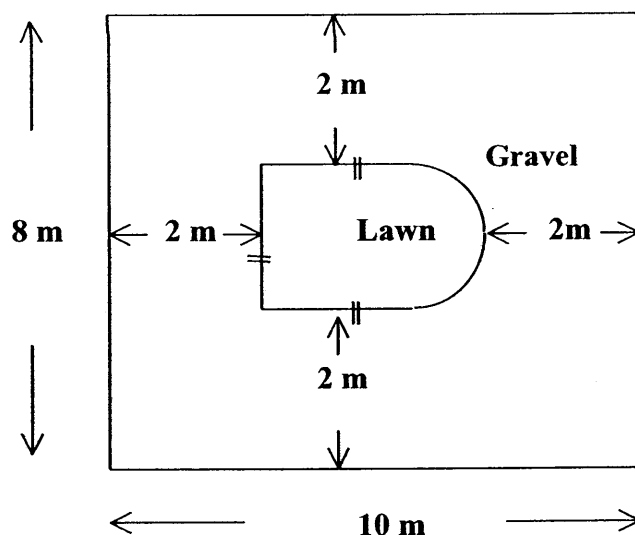
(c) A skirt is priced at \$82.50, including GST of 10%. What was the price of the skirt prior to adding GST? 1

(d) Evaluate  $(5.9 \times 10^{55})^2$  in scientific notation, correct to 2 significant figures 2

(e) Find the exact value of  $81^{-\frac{1}{4}} \div 2^{-2}$  1

(f) Express  $0.\dot{1}0\dot{2}$  as a fraction in it's simplest form. 2

(g) The Backyard Force TV team is renovating a backyard to the plan shown below. Calculate the area of lawn in the backyard (correct to 1 decimal place) 3

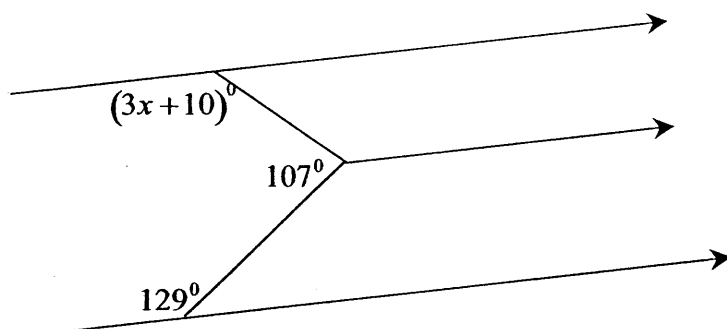


| Question 2 | Algebra and Surds (Start work on a new page)                           | Marks |
|------------|------------------------------------------------------------------------|-------|
| (a)        | Simplify $4 - 6(m - 2)$                                                | 1     |
| (b)        | If $S = \frac{a}{1-r}$ , find $S$ when $a = 0.8$ and $r = \frac{1}{2}$ | 1     |
| (c)        | Factorise completely $4a^2 - b^2 + 2a - b$                             | 2     |
| (d)        | Simplify $\sqrt{75} + \sqrt{48}$                                       | 2     |
| (e)        | Express $\frac{2\sqrt{3}-1}{5-\sqrt{3}}$ with a rational denominator.  | 2     |
| (f)        | Simplify $\frac{2}{x(x-3)} - \frac{1}{x}$                              | 2     |
| (g)        | Find the exact value of $\frac{t^3}{5} - t^2 + 1$ when $t = -3$        | 2     |

| Question 3 | Equations (Start work on a new page)                                 | Marks |
|------------|----------------------------------------------------------------------|-------|
| (a)        | Solve:                                                               | 1     |
|            | i. $3(2x - 3) = 4(x - 2)$                                            | 1     |
|            | ii. $\frac{5x}{2} - \frac{2x}{3} = 2$                                | 2     |
|            | iii. $x^2 - 3 = 3x + 1$                                              | 2     |
| (b)        | Mark on the number line the values of $x$ for which $ x - 1  \leq 2$ | 2     |
| (c)        | Find the values of $a$ and $b$ such that :                           |       |
|            | $\begin{cases} 2a - 3b = 7 \\ a - 4b = 1 \end{cases}$ and            | 2     |
| (d)        | Solve for $x$ : $(x + 1)^2 = 6$ , leaving your answer in exact form. | 3     |

**Question 4**      **Geometry** (Start work on a new page)

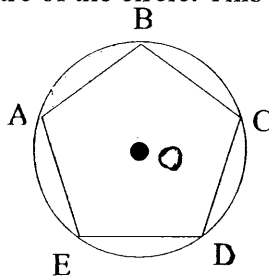
- (a) Find the value of  $x$  in the following diagram.



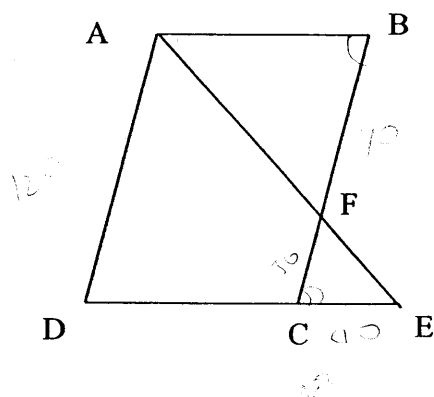
2

Marks

- (b) ABCDE is a regular pentagon where ABCDE lie on the circumference of a circle of diameter 32 cm. O is the centre of the circle. This is shown below.



- |       |                                                                                       |   |
|-------|---------------------------------------------------------------------------------------|---|
| (i)   | Find the length of OA.                                                                | 1 |
| (ii)  | Find the angle AOB.                                                                   | 1 |
| (iii) | Find the angle OAB.                                                                   | 1 |
| (iv)  | Find the length of the side of the pentagon, giving your answer to one decimal place. | 2 |
| (v)   | Find the area of the pentagon, giving your answer to one decimal place.               | 1 |
| (vi)  | Find the area of the sector AOB, giving your answer to one decimal place.             | 1 |
- (c) ABCD is a parallelogram with DC produced to E, where  $AD = 120\text{mm}$ ,  $CE = 40\text{mm}$  and  $BF = 70\text{mm}$ .
- (i) Show that  $\triangle ABF$  is similar to  $\triangle ECF$
- (ii) Find the length of AB



**Question 5**

**Functions and Graphs** (Start on a new page)

**Marks**

(a) A function  $f(x)$  is defined as:  $f(x) = \begin{cases} 2-x & \text{for } x < -3 \\ 5 & \text{for } -3 \leq x < 0 \\ x^2 - 1 & \text{for } x \geq 0 \end{cases}$

(i) Calculate the value of (a)  $f(-10)$

1

(β)  $f(-3)$

1

(γ)  $f(a^2)$

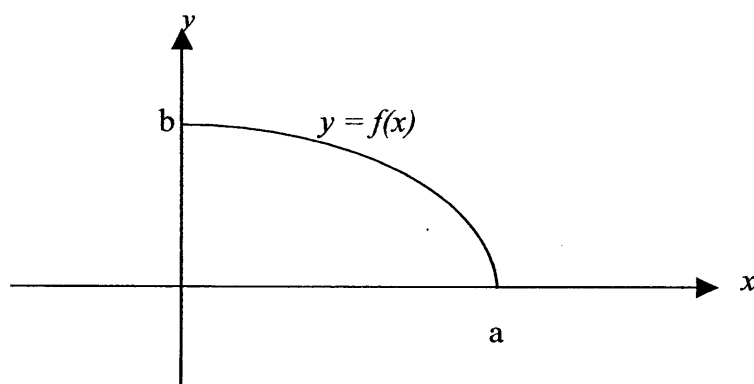
1

(ii) Sketch the graph of  $y = f(x)$  over the domain  $-5 \leq x \leq 2$ , showing all essential features including any intercepts with the  $x$  and  $y$  axes.

2

(b) The diagram below shows part of the graph of the function  $y = f(x)$ . You are told that it is an odd function. Copy the diagram and complete the graph of the function.

1



(c) Solve simultaneously:

2

$$\begin{cases} y = 4x + 3 \\ y = x^3 + 3 \end{cases}$$

(d) (i) Sketch the curve  $y = x^2 - 2x$ , clearly showing all intercepts.

1

(ii) What is the minimum value of this function?

1

(e) Sketch the graph of the function  $y = \frac{1}{x+1}$  and state the domain and the range of the function.

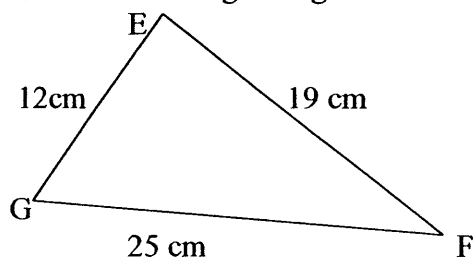
2

Question 6

Trigonometry (Start work on a new page)

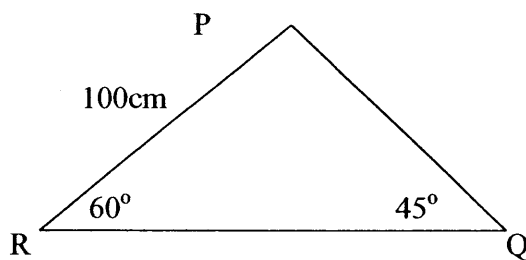
- (a) Find the size of the largest angle in the triangle below, correct to the nearest degree.

2



NOT TO SCALE

- (b) In  $\triangle PQR$  below:



- (i) Show that the length of PQ is  $50\sqrt{6}$  cm. 2
- (ii) Hence or otherwise, find the area of the triangle PQR. 2  
(correct to 1 decimal place)
- (c) Find all values of  $\theta$  such that  $\cos\theta = \frac{-1}{2}$  and  $0^\circ \leq \theta \leq 360^\circ$  2

- (d) (i) Show that:  $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$  1

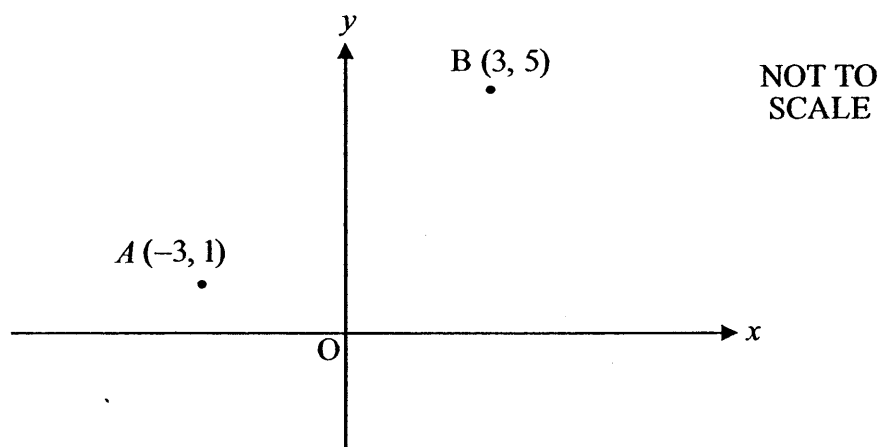
- (ii) Hence or otherwise, solve: 3

$$\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} = 1, \quad 0^\circ \leq \theta \leq 360^\circ$$

**Question 7**      **Straight Line Graphs** (*Start work on a new page*)

- (a) Sketch the graph of  $y = -x + 2$  on a set of axes, showing any  $x$  and  $y$  intercepts.

2



- (b) The diagram shows the points  $A(-3, 1)$  and  $B(3, 5)$  on the Cartesian number plane.

Copy or trace this diagram onto your writing page.

- (i) Show that the equation of  $AB$  is  $2x - 3y + 9 = 0$ .
- (ii) Show that the point  $C$ , which is the midpoint of  $AB$  is the  $y$ -intercept of  $AB$ .
- (iii) Calculate the perpendicular distance from the point  $D(2, 0)$  to the line  $AB$  and mark the point  $D$  on your diagram.
- (iv) The point  $E$ , lies on the line  $y = -1$  and the line  $BE$  is perpendicular to the line  $AB$ . Show that  $E$  has the coordinates  $(7, -1)$  and mark point  $E$  on your diagram.
- (v) Show that  $BCDE$  is a trapezium.
- (vi) Find the area of  $BCDE$ .

2

1

2

2

1

2

|                   |                                                                  |              |
|-------------------|------------------------------------------------------------------|--------------|
| <b>Question 8</b> | <b>Introductory Calculus</b> ( <i>Start work on a new page</i> ) | <b>Marks</b> |
|-------------------|------------------------------------------------------------------|--------------|

(a) Differentiate with respect to  $x$ .

(i)  $2x^4 + 7x^2 + 7$

1

(ii)  $x^{\frac{1}{3}}$

1

(iii)  $\sqrt{x^2 + 4}$

1

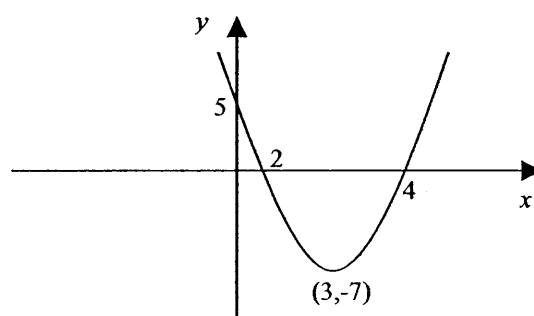
(iv)  $\frac{6x^2 - 7}{3 - 2x}$

2

(b) The graph shown below represents  $f(x)$

Sketch a graph to show the general shape of  $f'(x)$ .

1



(c) Find the equation to the normal to the curve  $y = x^2 + x$  at the point where  $x = 1$ .

2

(d) A tangent is drawn to the parabola  $y = x^2 - 4x$  at the point P. The tangent has a gradient of 6.

(i) Show that the coordinates of P are (5, 5).

2

(ii) Find the equation of the tangent at P.

1

(iii) Find the gradient of the normal at P.

1

**Question 9**

**Quadratic Functions** (*Start work on a new page*)

**Marks**

- |       |                                                                                                  |   |
|-------|--------------------------------------------------------------------------------------------------|---|
| (a)   | Find the value of $k$ for which $x^2 - (k-1)x - k = 0$ has equal roots                           | 2 |
| (b)   | Solve for $x$ : $x^6 - 7x^3 - 8 = 0$                                                             | 2 |
| (c)   | Find the values of $P$ , $Q$ and $R$ if $3x^2 + 5x - 1 \equiv P(x+1)^2 + Q(x+1) + R$             | 3 |
| (d)   | For what values of $x$ is $x^2 + 5x > 0$ ?                                                       | 2 |
| (e)   | For the quadratic expression: $2x^2 - x + 3$ which has zeros when $x = \alpha$ and $x = \beta$ . |   |
| (i)   | Show that the expression is positive definite.                                                   | 1 |
| (ii)  | Find the value of $\alpha\beta$ .                                                                | 1 |
| (iii) | Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ .                                         | 1 |

| Question 10 | Locus and the Parabola (Start work on a new page)                                                                                                                    | Marks |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| (a)         | For the parabola $16y = x^2$ write down the coordinates of the focus and the equation of the directrix.                                                              | 2     |
| (b)         | The points P and Q have the coordinates $(-1, 0)$ and $(3, 3)$ respectively. If PR is perpendicular to QR, show that the locus of R is $x^2 + y^2 - 2x - 3y - 3 = 0$ | 2     |
| (c)         | Find the equation of the parabola which has its vertex at $(2, 0)$ and its directrix is given by $x = 5$ .                                                           | 2     |
| (d)         | By expressing $y = \frac{x^2}{-16} + \frac{6x}{-16} - \frac{7}{-16}$ in the form $(x - h)^2 = 4a(y - k)$ , find:                                                     | 6     |
|             | (i) The focal length                                                                                                                                                 |       |
|             | (ii) The coordinates of the vertex                                                                                                                                   |       |
|             | (iii) The equation of the axis                                                                                                                                       |       |
|             | (iv) The equation of the directrix                                                                                                                                   |       |
|             | (v) Sketch the graph                                                                                                                                                 |       |

End of paper

# Suggested Answers

## Task Mathematics Preliminary 2003

| Question 1                                                               |       |                                        |
|--------------------------------------------------------------------------|-------|----------------------------------------|
| a. -3                                                                    | 1     | 2 continued                            |
| b. 0.78                                                                  | 2     | e. $(2\sqrt{3}-1)(5+\sqrt{3})$         |
| c. 3.75                                                                  | 1     | 25-3                                   |
| d. $34.81 \times 10^{110}$ (without comma!)                              | 1     | $= \frac{10\sqrt{3}-5-\sqrt{3}+5}{22}$ |
| $= 3.481 \times 10^{111}$                                                | 1/2   | $= \frac{9\sqrt{3}+1}{22}$             |
| $= 3.5 \times 10^{111}$ (2.55)                                           | 1/2   |                                        |
| e. $\frac{4}{3}$                                                         |       |                                        |
| f. $x=0.102102...$                                                       |       |                                        |
| $1000x=102.102...$                                                       | 1 1/2 | f. $\frac{2}{x(x-3)} - \frac{1}{x}$    |
| $999x=102$                                                               |       | $= \frac{2-(x-3)}{x(x-3)}$             |
| $x=\frac{102}{999}$                                                      |       | $= \frac{5-x}{x(x-3)}$                 |
| $= \frac{34}{333}$                                                       | 2     |                                        |
| g. Area = $4 \times 4 + \frac{1}{2} \times \pi \times 2^2$               | 2     |                                        |
| $= 16 + 2\pi$                                                            |       |                                        |
| $\div 22.3m^2$                                                           | 3     |                                        |
| Question 2                                                               |       |                                        |
| a. $4-6(m-2)$                                                            | 1     |                                        |
| $= 4-6m+12$                                                              |       |                                        |
| $= 16-6m$                                                                |       |                                        |
| b. $S=\frac{0.8}{1-\frac{1}{2}}$                                         | 1     |                                        |
| $= 1.6$                                                                  |       |                                        |
| c. $(2a-b)(2a+b) + (2a-b)$                                               | 1     |                                        |
| $= (2a-b)(2a+b+1)$                                                       |       |                                        |
| d. $\sqrt{15} + \sqrt{48}$                                               | 1     |                                        |
| $= 5\sqrt{3} + 4\sqrt{3}$                                                |       |                                        |
| $= 9\sqrt{3}$                                                            |       |                                        |
| e) $\frac{2\sqrt{3}-1}{5-\sqrt{3}} \times \frac{5+\sqrt{3}}{5+\sqrt{3}}$ | 1     |                                        |

| Question 2 cont.                      |             | Task |
|---------------------------------------|-------------|------|
| b. $x-1=2$                            | $-x-1=2$    |      |
| $x=3$ (i)                             | $-x+1=2$    |      |
| $ x-1  \leq 2$                        | $x=-1$ (ii) |      |
| test $x=0$                            |             |      |
| $-1 \leq 1 \leq 2$ ✓ true             |             |      |
| c. $2a-3b=7$                          | $2a-8=2$    |      |
| $2a-8b=2$                             | $2a=10$     |      |
| $5b=5$                                | $a=5$       |      |
| $b=1$                                 |             |      |
| d. $x+1=\pm\sqrt{6}$                  |             |      |
| $x=-1\pm\sqrt{6}$                     |             |      |
| $x^2+2x+1=6$                          |             |      |
| $x^2+2x-5=0$                          |             |      |
| $x=\frac{-2\pm\sqrt{4+4\times 5}}{2}$ |             |      |
| $=\frac{-2\pm\sqrt{24}}{2}$           |             |      |
| $=\frac{-2\pm 2\sqrt{6}}{2}$          |             |      |
| $=-1\pm\sqrt{6}$                      |             |      |
| <u>Question 4</u>                     |             |      |
| a) $3x+10+107+129=360^\circ$          |             |      |
| $3x=114$                              |             |      |
| $x=38^\circ$                          |             |      |
| b) $OA=16cm$                          |             |      |
| $\frac{360}{5}=72^\circ$              |             |      |
| $\angle AOB=72^\circ$                 |             |      |
| $\angle COB=54^\circ$                 |             |      |
| <u>Question 5</u>                     |             |      |
| i. a) $f(-10)=2-(-10)$                |             |      |
| $=12$                                 |             |      |
| b) $f(-3)=5$                          |             |      |
| c) $f(a^2)=(a^2)^2-1$                 |             |      |
| $=a^4-1$                              |             |      |
| d) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| e) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| f) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| g) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| h) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| i) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| j) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| k) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| l) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| m) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| n) $f(a^2)=a^4-1$                     |             |      |
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| o) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| p) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| q) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| r) $f(a^2)=a^4-1$                     |             |      |
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| s) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| t) $f(a^2)=a^4-1$                     |             |      |
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| u) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| v) $f(a^2)=a^4-1$                     |             |      |
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| w) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| x) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| y) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| z) $f(a^2)=a^4-1$                     |             |      |
| $=a^4-1$                              |             |      |
| aa) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ab) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ac) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ad) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ae) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| af) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ag) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ah) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ai) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| aj) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ak) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| al) $f(a^2)=a^4-1$                    |             |      |
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| $=a^4-1$                              |             |      |
| an) $f(a^2)=a^4-1$                    |             |      |
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| ao) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ap) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| aq) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ar) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| as) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| at) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| au) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| av) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| aw) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ax) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ay) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| az) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ba) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bb) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bc) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bd) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| be) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bf) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bg) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bh) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bi) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bj) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bk) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bl) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bm) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bn) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bo) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bp) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bq) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| br) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bs) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bt) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bu) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bv) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bw) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bx) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| by) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| bz) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ca) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cb) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cc) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cd) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ce) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cf) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cg) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ch) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ci) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cj) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ck) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cl) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cm) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cn) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| co) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cp) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cq) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cr) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cs) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ct) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cu) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cv) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cw) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cx) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cy) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| cz) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| da) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| db) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dc) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dd) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| de) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| df) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dg) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dh) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| di) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dj) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dk) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dl) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dm) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dn) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| do) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dp) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dq) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dr) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ds) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dt) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| du) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dv) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dw) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dx) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dy) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| dz) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ea) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| eb) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ec) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ed) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ee) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ef) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| eg) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| eh) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ei) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ej) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ek) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| el) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| em) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| en) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| eo) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ep) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| eq) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| er) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| es) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| et) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| eu) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ev) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ew) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ex) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ey) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |
| ez) $f(a^2)=a^4-1$                    |             |      |
| $=a^4-1$                              |             |      |

# Task Maths Prelim. Suggested answers 2008

## Question 6

c.  $y = 4x + 3$   
 $y = 8x^3 + 3$

$x^3 + 3 = 4x + 3$

$x^3 - 4x = 0$

$x(x+2)(x-2) = 0$

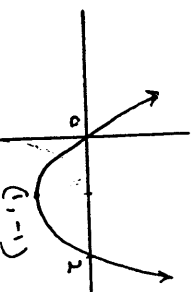
$x = 0, \pm 2$

$x = -2, y = -5$

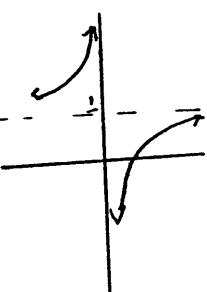
$x = 0, y = 3$

$x = 2, y = 11.$

d.  $y = x^2 - 2x$   
 $= x(x-2)$



min value is -1



Domain all  $x, y \neq -1$   
 Range all  $y$ , except  $y \neq 0$

## Question 7

a.  $\cos \theta = \frac{12^2 + 19^2 - 25^2}{2 \times 12 \times 19}$

$\theta = 105^\circ$

i)  $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$

$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$

$= \frac{1}{\sin \theta \cos \theta}$

$= \frac{1}{\frac{1}{2} \sin 2\theta}$

ii)  $(1 + \frac{\cos \theta}{\sin \theta}) \div \frac{1}{\sin \theta} = \frac{1}{\cos \theta} \times \frac{\sin \theta}{1} = \frac{\sin \theta}{\cos \theta} = \tan \theta$

$= \frac{\sin \theta}{\cos \theta} \times \frac{1}{1} = \frac{\sin \theta}{\cos \theta} = \tan \theta$

$\therefore \cos \theta = 1$

$\theta = 0, 360^\circ$

Sorry about order!

$\frac{x}{\sin 60} = \frac{100}{\sin 45}$

$\frac{x}{\sqrt{3}/2} = \frac{100}{1/\sqrt{2}}$

$2x = 100\sqrt{6}$

$x = 50\sqrt{6}$

ii)  $A = \frac{1}{2} \times 100 \times 50\sqrt{6} \sin 75^\circ$

$\approx 5915.1 \text{ cm}^2$

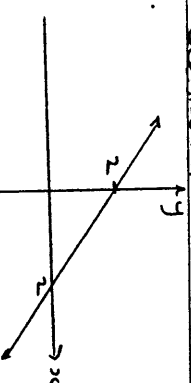
c)  $\theta = 60^\circ$  in the 2nd/3rd quad

$\therefore \theta = 120^\circ, 240^\circ$

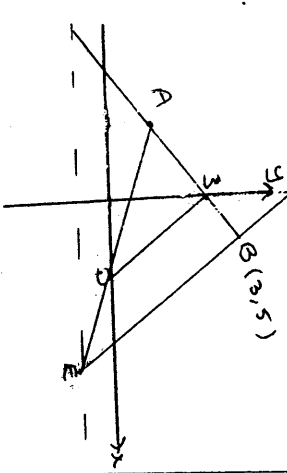
## Question 7

Task

a.



b.



i)  $m = \frac{2}{3}$

$y - 1 = \frac{2}{3}(x + 3)$

$3y - 3 = 2x + 6$

$2x - 3y + 9 = 0$

ii)  $m(x, y) = (\frac{2}{3}, \frac{1}{3}) = (0, 3)$

$2x - 3y + 9 = 0$

at  $x = 0$

$0 - 3y + 9 = 0$

$\therefore y = 3.$

$\therefore$  the midpoint of AB is also

the y-int. of the line AB

iii)  $AD = \frac{|2 \times 1 - 3 \times 0 + 9|}{\sqrt{4 + 9}}$

$= \frac{11}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$

iv)  $m_2 = -\frac{3}{2} = \sqrt{13} \text{ units}$

$y - 5 = -\frac{3}{2}(x - 3)$

$2y - 10 = -3x + 9$

$3x + 2y - 19 = 0$

at  $y = -1, 3x - 2 - 19 = 0$

v)  $m \text{ of } CD = -\frac{3}{2}$

$m \text{ of } BE = -\frac{3}{2}$

$\therefore CD \parallel BE$

$\therefore BCDE$  is a quad. & pair of parallel lines &  $\therefore$  a trapezium

vi)  $BE = \sqrt{16 + 36} = \sqrt{52} = \sqrt{4 \times 13} = 2\sqrt{13}$

$BC = \sqrt{13}$

$\therefore A = \frac{1}{2} \times \sqrt{13} (\sqrt{13} + 2\sqrt{13})$

$= \frac{\sqrt{13}}{2} \times 3\sqrt{13}$

$= \frac{39}{2} \text{ u}^2$

## Question 8

i)  $8x^3 + 14x$

ii)  $\frac{1}{3} x$

iii)  $\frac{1}{2} (x^2 + 4) \times 2x$

$= \frac{x}{\sqrt{x^2 + 4}}$

iv)  $f(x) = \frac{3x^2 - 7}{3 - 2x}$

$f'(x) = \frac{(3 - 2x)(12x - 2(6x^2 - 7))}{(3 - 2x)^2}$

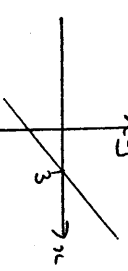
$= \frac{36x - 24x^2 + 12x^2 - 14}{(3 - 2x)^2}$

$= \frac{-12x^2 + 36x - 14}{(3 - 2x)^2}$

$= \frac{-12x^2 + 36x - 14}{(3 - 2x)^2}$

$= \frac{-12x^2 + 36x - 14}{(3 - 2x)^2}$

b.



Question 8 cont.

c.  $y = x^2 + 2x$

$$y' = 2x + 1$$

$$\text{at } x=1, y=2$$

$$\therefore m_1 = 3$$

$$m_2 = -\frac{1}{3}$$

$$y - 2 = -\frac{1}{3}(x - 1)$$

$$3y - 6 = -x + 1$$

$$\therefore x + 3y - 7 = 0.$$

$$\text{or } y = -\frac{1}{3}x + \frac{7}{3}.$$

d.  $y = x^2 - 4x$

$$y' = 2x - 4$$

$$\text{at } y' = 0$$

$$2x - 4 = 0$$

$$2x = 4$$

$$x = 2, y = 25 - 20 = 5$$

$$\therefore P \text{ is } (5, 5)$$

$$y - 5 = 6(x - 5)$$

$$y - 5 = 6x - 30$$

$$y = 6x - 25$$

iii)  $m_2 = -\frac{1}{6}$

Question 9

a)  $x^2 - (x-1)x - k = 0$

$$\Delta = 0$$

$$(k-1)^2 + 4k = 0.$$

$$k^2 - 2k + 1 + 4k = 0$$

$$k^2 + 2k + 1 = 0$$

$$(k+1)^2 = 0$$

$$\therefore k = -1.$$

Question 10 cont.

b.  $x^2 - 7x^3 - 8 = 0$

$$(x^3 - 8)(x^2 + 1) = 0$$

$$\therefore x^3 = 8, -1$$

$$x = 2, x = -1$$

c.  $3x^2 + 5x - 1 = P(x+1)^2 + Q(x+1) + R$

$$= Px^2 + 2Px + P + Qx + Q + R$$

$$= Px^2 + (2P+Q)x + P+Q+R$$

$$\therefore P=3, 2P+Q=5, P+Q+R=-1$$

$$Q=-1, R=-3$$

d.  $x^2 + 5x > 0$

$$x(x+5) > 0$$

$$x < -5, \text{ or } x > 0$$

e)  $\Delta = 1 - 11, a = 2$

$$= -10$$

since  $\Delta < 0$  &  $a > 0$ ,

$\Rightarrow x$  is definite

ii)  $aP = \frac{2}{3}$  iii)  $\frac{1}{x} + \frac{1}{y}$

$$= \frac{3}{2}$$

$$= \frac{d+g}{2} = \frac{1}{3}$$

$$= \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$$

$$= \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$$

$$= \frac{1}{12} \cdot \frac{1}{2} = \frac{1}{24}$$

Question 10

a) focus  $(0, 4)$

directrix  $y = -4$

b)  $PR = \frac{y-0}{x+1}$  (t)  $QR = \frac{y-3}{x-3}$  (t)

$$\frac{y}{x+1} = -\frac{(x-3)}{y-3}$$

$$y(y-3) = -(x-3)(x+1)$$

$$y^2 - 3y = -(x^2 - 3x + x - 3)$$

$$y^2 - 3y = -x^2 + 2x + 3$$

$$\therefore x^2 + y^2 - 2x - 3y - 3 = 0.$$

Question 10 cont.

c)  $(y-k)^2 = 4a(x-h)$

$$(y-0)^2 = -4 \times 3(x-2)$$

$$y^2 = -12(x-2)$$

$$y^2 = -12(x-2)$$

$$y^2 = -12(x-2)$$

d)  $-16y = x^2 + 6x - 7$

$$x^2 + 6x + 9 = -16y + 7 + 9$$

$$(x+3)^2 = -16(y-1)$$

$$(x+3)^2 = -16(y-1)$$

$$i) a = 4 \text{ (not -4)}$$

$$ii) (-3, 1)$$

$$iii) x = -3$$

$$iv) y = 5$$

